INNOVATIVE MINING SYSTEMS
Technology life cycle curve

By

Prof. Santosh Kumar Behera, Assistant Professor,

Mining Engineering

Indian Institute of Technology (Indian School of Mines), Dhanbad

S – Curve of Technological Progress

- Technologies usually follow a certain pattern
- Managing technologies require a deep understanding of the life cycle of the technology, product, process and system
- The course of technology's performance is typically depicted through an S – Curve.
- Technology performance can be depicted (on the Y axis) by any attribute such as density (of populating a chip) or aircraft speed in m.p.h. etc.

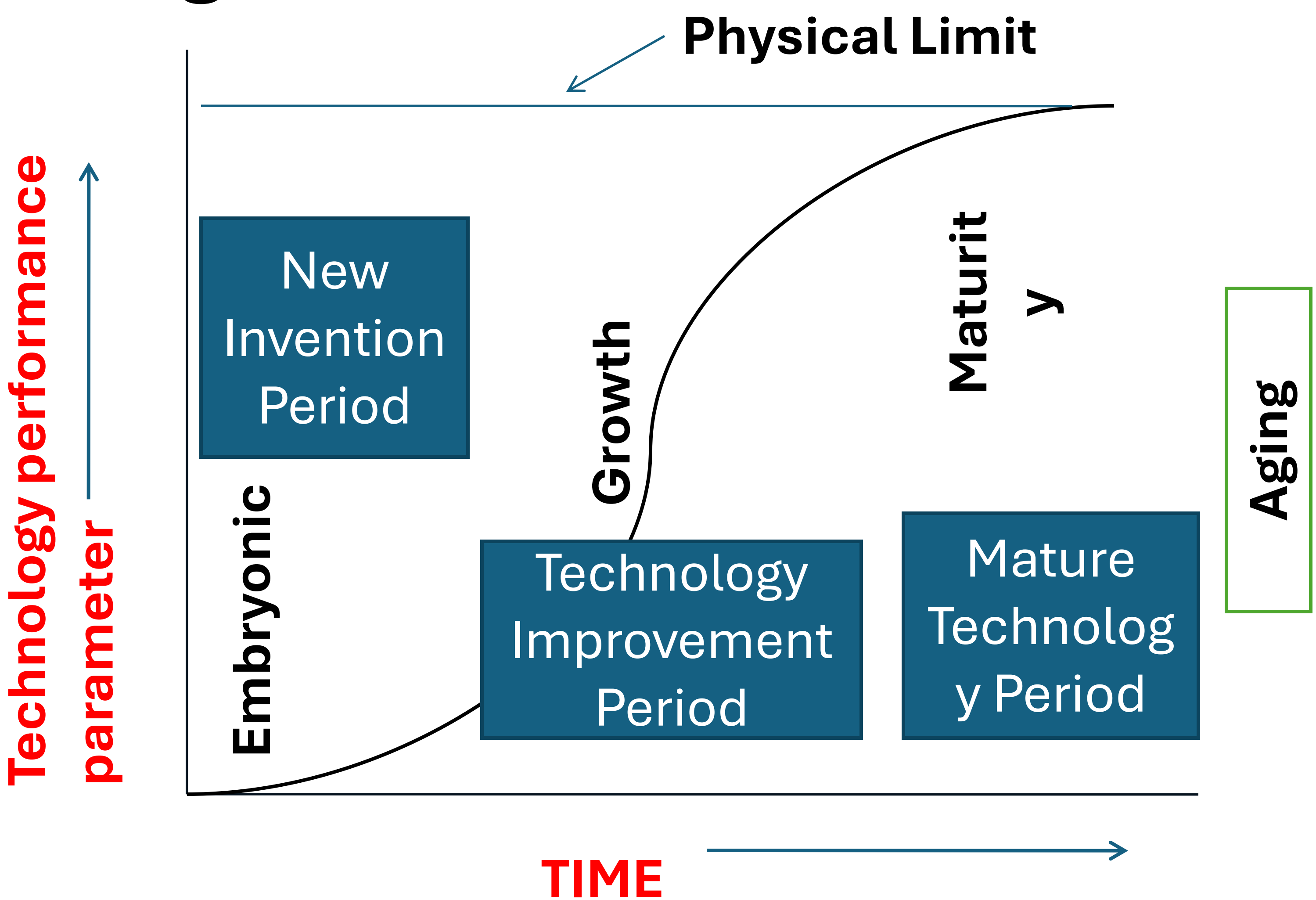
S – Curve of Technological Progress

- Technology progresses through a three stage technology life cycle (TLC)

1) The new invention period or Embryonic stage
2) Technology improvement period or growth stage (exponential growth)
3) Mature technology period or tapering off stage (when further progress is prohibitively expensive or too time consuming; hence not worthwhile.) .

- New or better performing technology emerges in it's place. This also exhibits the same 3 stage cycle. This cycle gets repeated over time.

S – Curve of Technological Progress



S – Curve of Technological Progress

Example:

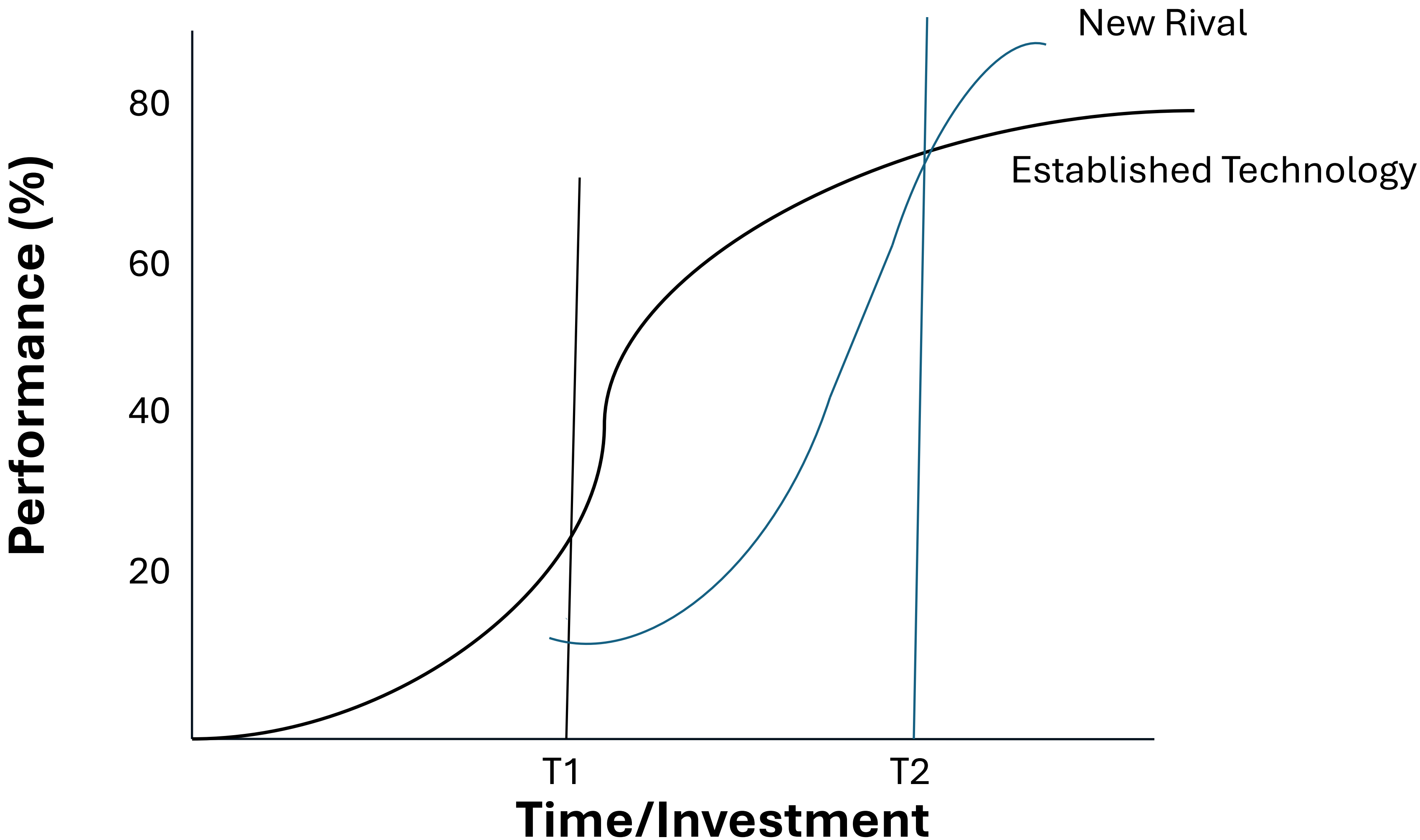
Vacuum Tube technology was limited by the tube's size and the power consumption of the heated filament. Both of these factors were natural barriers which electronic engineers could not overcome. Hence there was saturation in this technology

The arrival of the solid state technology –the transistor started a new technology life cycle. The transistor permitted electron conduction in solid state material, changed the physical barriers of size and power.

S – Curve of Technological Progress

- When a technology reaches its natural limits it becomes a mature technology vulnerable to substitution or obsolescence.
- The value of the limit reached may differ depending on the basic nature of the technology and the time/cost devoted to it’s development.
- The new technology usually has a higher limit of performance compared to the older one and may improve the progression of the older technology

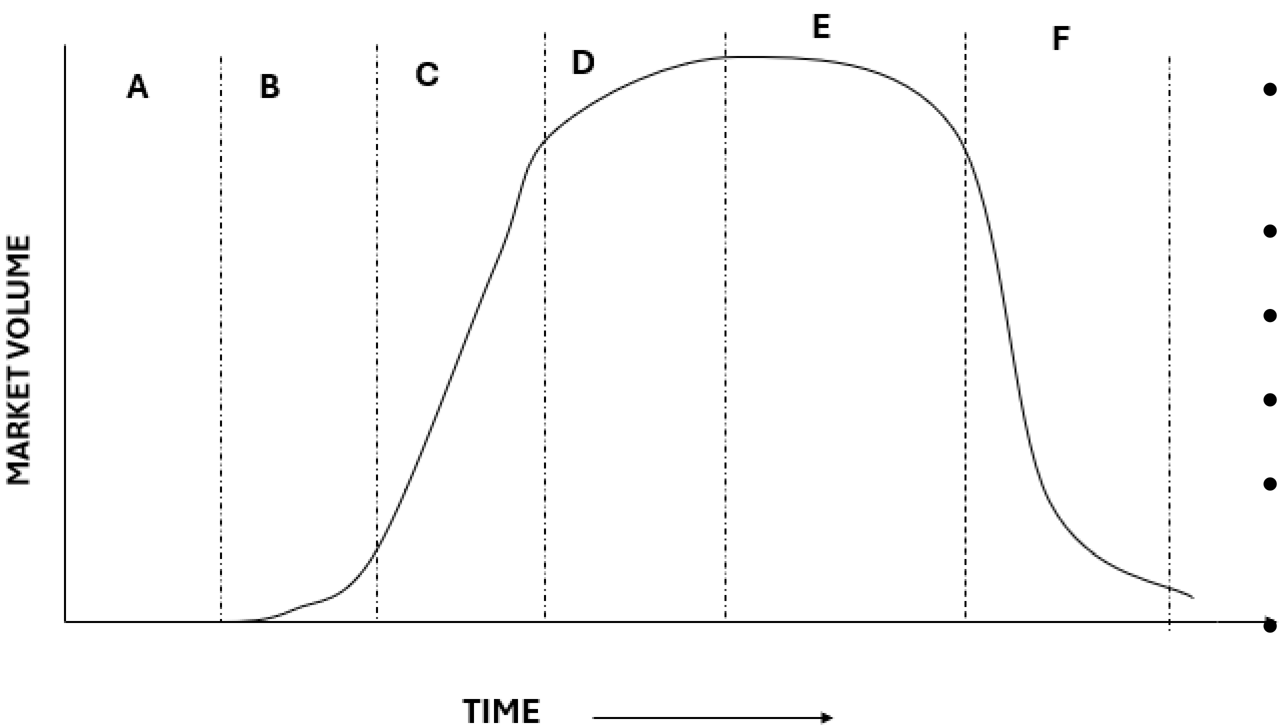
S – Curve – an established technology and a new rival.



S – Curve of Technological Progress

- Example:
- Metals had reached a limit of ‘operating temperature’ because of which IC engines had reached a saturation in performance.
- However, with the invention of engineering high performance ceramics , the limits were substantially raised which in turn enabled the IC engines to once again grow in performance parameter.

MARKET GROWTH AT DIFFERENT STAGES OF THE TECHNOLOGY LIFE CYCLE



- A- Technology development
- B- Application launch
- C- Application growth
- D-Mature Technology
- E- Technology Substitution
- F-Technology obsolescence

S – Curves of Multiple Technologies

- A major technology can have a number of sub technologies associated with it and these sub-technologies have their won S Curves.
- Example: The Personal Computing is a mother technology for which the micro-processor is an essential part and it has it’s own technology base and S Curve. (8088, 286, 386, 486, pentium)

Technology and Market interaction

- The presence of a market or the creation of a new market is the reward for a technological innovation.
- Science technology push: Science creates a base for technology developments which in turn creates new markets

Science-Technology Push

Technology	Scientific Discovery
Nuclear Energy	Based on Einstein's 1905 paper, which establishes the equivalence of mass and energy
Transistors	Based on A.H.Wilson's 1931 theory of semi-conductors
Electronics	Based on Maxwell's theory of electromagnetism, developed in 1880s
Genetic Engineering	Based on Watson and Crick'w 1952 discovery of the structure of DNA (double-helix structure of DNA). This discovery explained how genetic information is stored, copied, and passed from one generation to the next, marking a revolution in biology and medicine.

Market Pull

- Technology development is also stimulated by Market Pull. Market pull is usually stimulated by consumers. E.G. COVID, a new disease at that time, resulted in invention of drugs through research.
- Consumers may or may not know that a new technology is being developed or is possible to be developed. E.g Consumers never knew that wireless technology or Electricity was possible.

Combined Market Pull-Technology Push

- Munro and Noori (1988) proposed that commitment to technology adoption is dependent on an integrated approach to technology push and market pull combined with management's attitude towards technology and the firm's technical and financial resources.